Sahaj Bhadja

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Summary

Junior Software Engineer with 2+ years building **data-intensive production systems** using **Python, TypeScript, SQL, AWS**, including APIs, data pipelines, secure web applications, observability, automated testing, and AI-assisted development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a **multi-tenant data platform** with normalized schemas, row-level isolation, and indexed SQL queries, reducing dashboard latency from 1.2 seconds to 350 milliseconds while supporting 500+ users.
- Built API and data pipelines across FastAPI, SQL Server, PostgreSQL, and cloud storage, processing 5K+ monthly jobs with **validation, reconciliation, monitoring, and CI/CD** while maintaining 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built **React and Flask production features** for uploads, workflow history, and feedback, serving 150+ instructors and 10K+ daily requests while translating user requirements into reliable interfaces and APIs.
- Implemented multithreaded processing and automatic fallbacks for 1K+ files per batch, reducing turnaround 60% while validating changes through **unit testing, debugging, Git reviews, and CI/CD** releases safely.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for autonomous-driving planning, improving trajectory results 18% while processing 100K+ simulation logs through reproducible **Python data and evaluation pipelines** for controlled performance experiments.
- Integrated HD-map geometry, lane context, and agent trajectories into a unified model, improving closed-loop stability 15% while documenting **software behavior, performance requirements, and evaluation results** across simulation baselines.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built **AWS Glue and S3 data pipelines** for JSON, CSV, and Parquet data, improving processing efficiency 68% while using object-oriented patterns to reduce API latency 30% and technical debt 20%.
- Automated 20+ **unit, integration, and regression tests** using PyTest, Mockito, and Jenkins across Windows, Linux, and macOS, collaborating through Agile reviews and Git workflows to improve production releases.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **React and Node.js web applications** with REST APIs, WebSockets, authentication, billing, permissions, and notifications, synchronizing 2M+ records across distributed locations through reliable real-time workflows for users daily.
- Owned eight services through implementation and deployment, adding **secure dependency checks, automated testing, monitoring, Docker, and Jenkins CI/CD** that cut release time 50% while sustaining 95%+ reliability.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C++, JSON, Bash
Data & APIs: FastAPI, Flask, Node.js, REST APIs, WebSockets, PostgreSQL, SQL Server, MySQL, MongoDB, Redis, Parquet, Data Modeling
Cloud & Systems: AWS, S3, AWS Glue, Azure, Docker, Kubernetes, Event-Driven Systems, Infrastructure as Code
AI & Applications: React.js, LLM APIs, AI Agents, RAG, Semantic Search, Embeddings, FAISS, PyTorch, Simulation Pipelines
Engineering: Git, GitHub Actions, CI/CD, Jenkins, PyTest, Mockito, Jest, Unit Testing, Integration Testing, Debugging, Secure Development, Monitoring, Observability, DevSecOps

Projects

RaceData Core | *Python, TypeScript, FastAPI, PostgreSQL, Parquet, AWS, Docker*

- Designed a **telemetry data platform** with relational and Parquet models, FastAPI ingestion, event-driven processing, validation, and cloud storage, exposing reliable APIs and dashboards for simulation and performance analysis.

SecureOps Console | *TypeScript, React, Node.js, AWS, GitHub Actions, Infrastructure as Code*

- Built a **production operations console** for deployment status, service metrics, security findings, and incident workflows, integrating event-driven APIs, automated validation, observability, and infrastructure changes through reviewable CI/CD pipelines.

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, Docker, Maven*

- Built an **AI-assisted database workspace** across Java and TypeScript modules, supporting 10+ database systems while accelerating schema exploration, SQL authoring, query validation, and cross-platform data management through Docker workflows.